

LAB 2: EXPLORATORY DATA ANALYSIS (EDA) ON A DATASET

#THEORY

This lab utilizes a dataset containing geometric features computed from digitized images of Fine Needle Aspiration (FNA) breast mass biopsies. The primary objective is to use Exploratory Data Analysis (EDA) to determine how cellular characteristics differentiate between Benign (B) and Malignant (M) tumors.

By inspecting data integrity, we ensure clean inputs by removing missing or duplicate values. Univariate distributions (histograms) reveal the physical limits of cell growth, while bivariate scatter plots visually isolate how size and structural irregularities cluster by diagnosis.

Finally, a correlation heatmap identifies multicollinearity among highly dependent geometric metrics (like radius and area), providing a foundational step for future predictive machine learning modeling.

#CONCLUSION

This lab successfully utilized an EDA pipeline to analyze nuclear morphometric data, confirming that malignant tumors distinctly cluster around higher values for size, concavity, and irregularity. Data cleaning verified dataset integrity, while distribution plots and the correlation heatmap exposed critical right-skewed profiles and feature redundancies. Ultimately, this quantitative analysis proves that computational cell profiling provides a clear, distinct signal for automating breast cancer diagnosis.